Wherobots

From Pixels to Patches

Pooling Strategies for Earth Embeddings

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Motivation

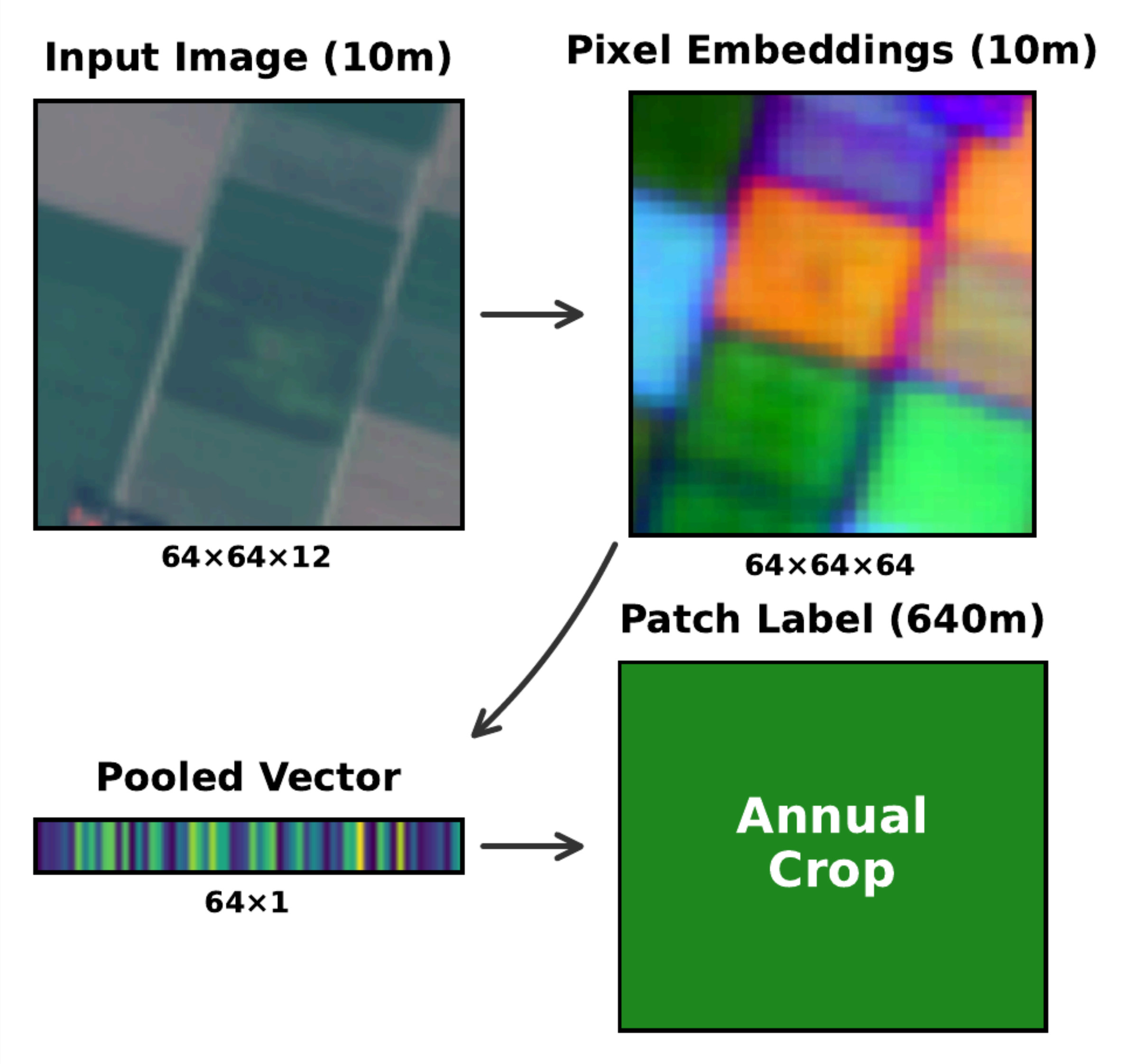
The resolution mismatch problem

Geospatial foundation models produce **pixel-level** embeddings (64–512 dims per pixel).

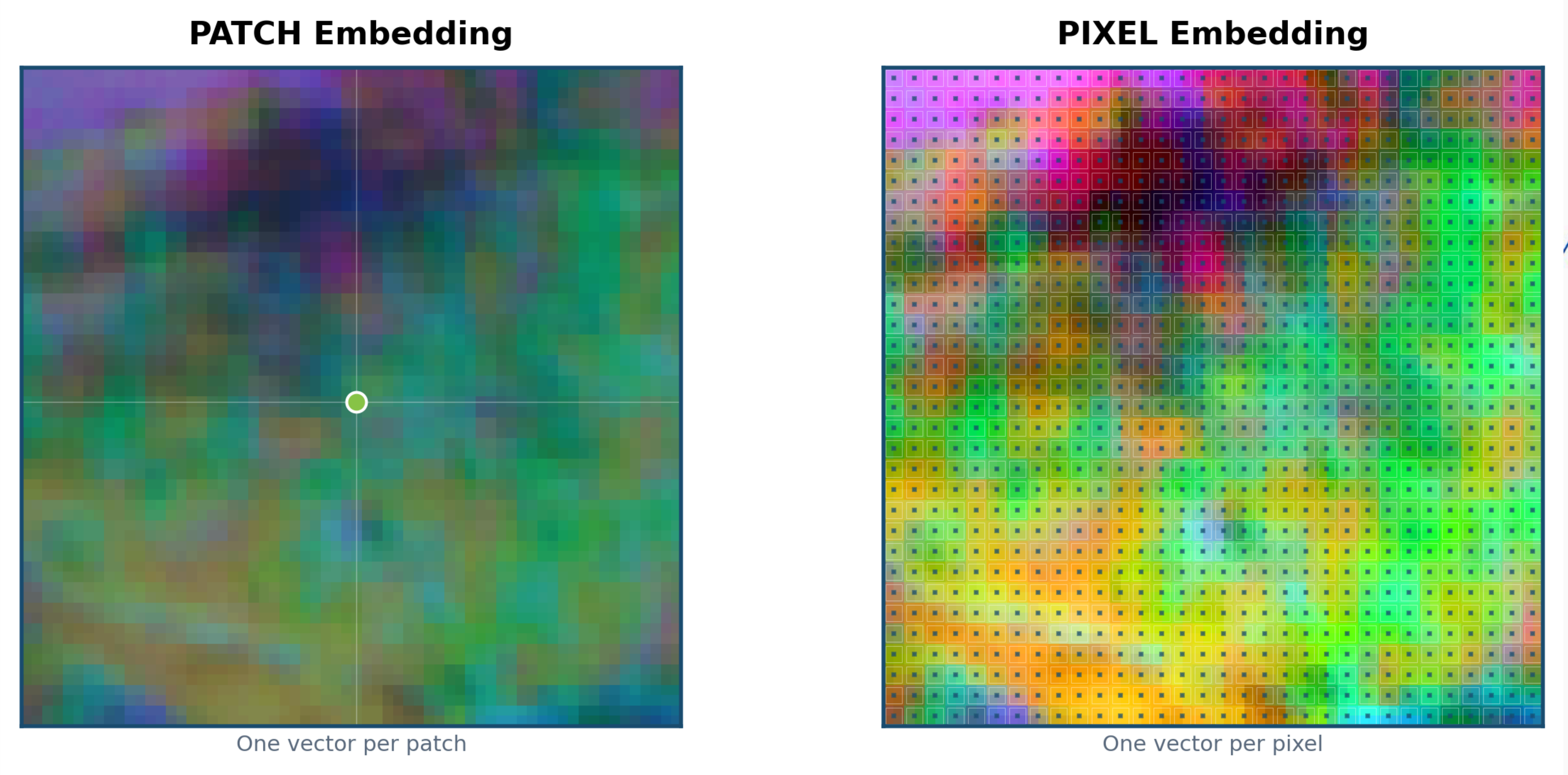
But downstream labels live at **patch / region** scale — land cover patches, crop fields, building polygons.

So we must pool. The question is *how*.

The default — mean pooling — drops accuracy by >10% under geographic shift.



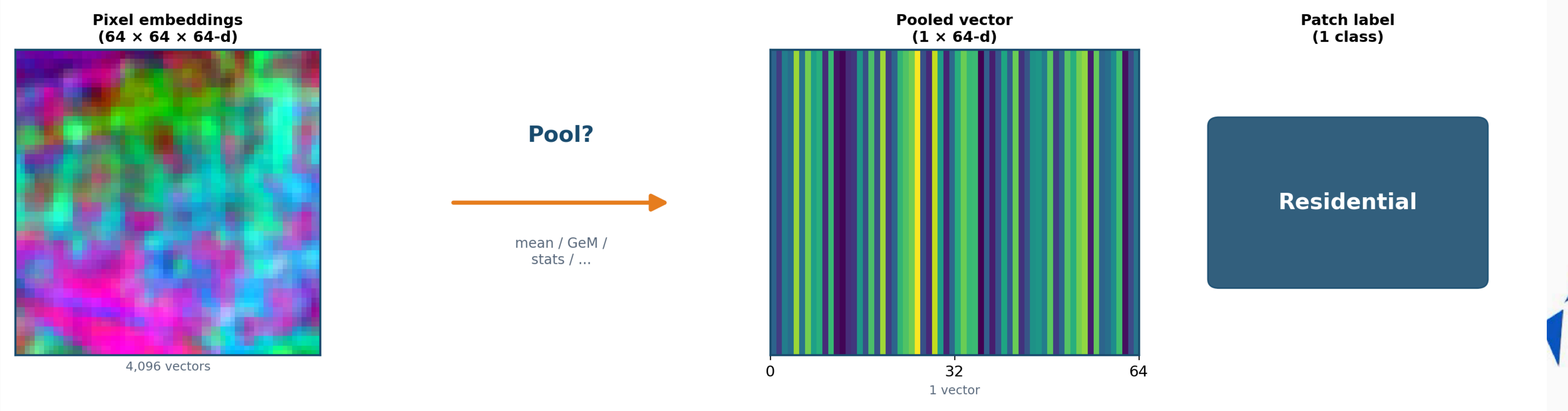
Patch vs. pixel embeddings



One vector per patch   One vector per pixel

Pixel-level models produce a vector *per pixel* — thousands of embeddings per patch that must be aggregated.

The pooling pipeline



PCA pseudo-RGB of actual AlphaEarth embeddings (64-d). Pooling compresses 4,096 vectors into one — the method matters.

Pooling matters in other domains

Image Retrieval

GeM pooling consistently outperforms mean pooling

Radenović et al., 2018

Audio / Video

Stats pooling (mean+std) summarizes variable-length sequences effectively

Miech et al., 2017

Object-Based IA

Bag of Visual Words uses region histograms for classification

Csurka et al., 2004

Yet pooling is **largely unexplored** for earth embeddings.

EuroSAT-Embed

EuroSAT-Embed: a pooling research benchmark

81,000 pixel-embedding GeoTIFFs from three foundation models, aligned to EuroSAT (27k patches, 10 land-cover classes).

AlphaEarth	OlmoEarth-Nano	Tessera
64-d, `int8`, 3.3 GB	128-d, `float32`, 40 GB	512-d, `float32`, 164 GB

Reproducible pooling research **without re-running** expensive GFM inference.

Evaluated on **random** (i.i.d.) and **spatial** (geographically disjoint) splits.

 · hf.co/datasets/isaaccorley/eurosat-embed

Pooling Methods

First-order pooling

Given $X \in \mathbb{R}^{H\times W\times D}$ with $N = HW$ pixels:

Mean (baseline)

$$\mu = \frac{1}{N}\sum_i x_i$$

Max — per-channel maximum

Generalized Mean (GeM)

$$\mathrm{GeM}_p = \left(\frac{1}{N}\sum_i {x_i}^p\right)^{1/p}$$

Center-weighted mean — Gaussian kernel

All preserve original dimensionality (1×).

Distributional and parametric pooling

Hypothesis: class signal lives in the *distribution*, not just the mean.

Distributional (training-free):

- Mean+Std, Mean+Max (2×)
- Median+IQR (2×)
- Stats [min/max/μ/σ] (4×)
- Percentiles [10..90] (5×)
- Covariance upper-tri

Parametric (need training data):

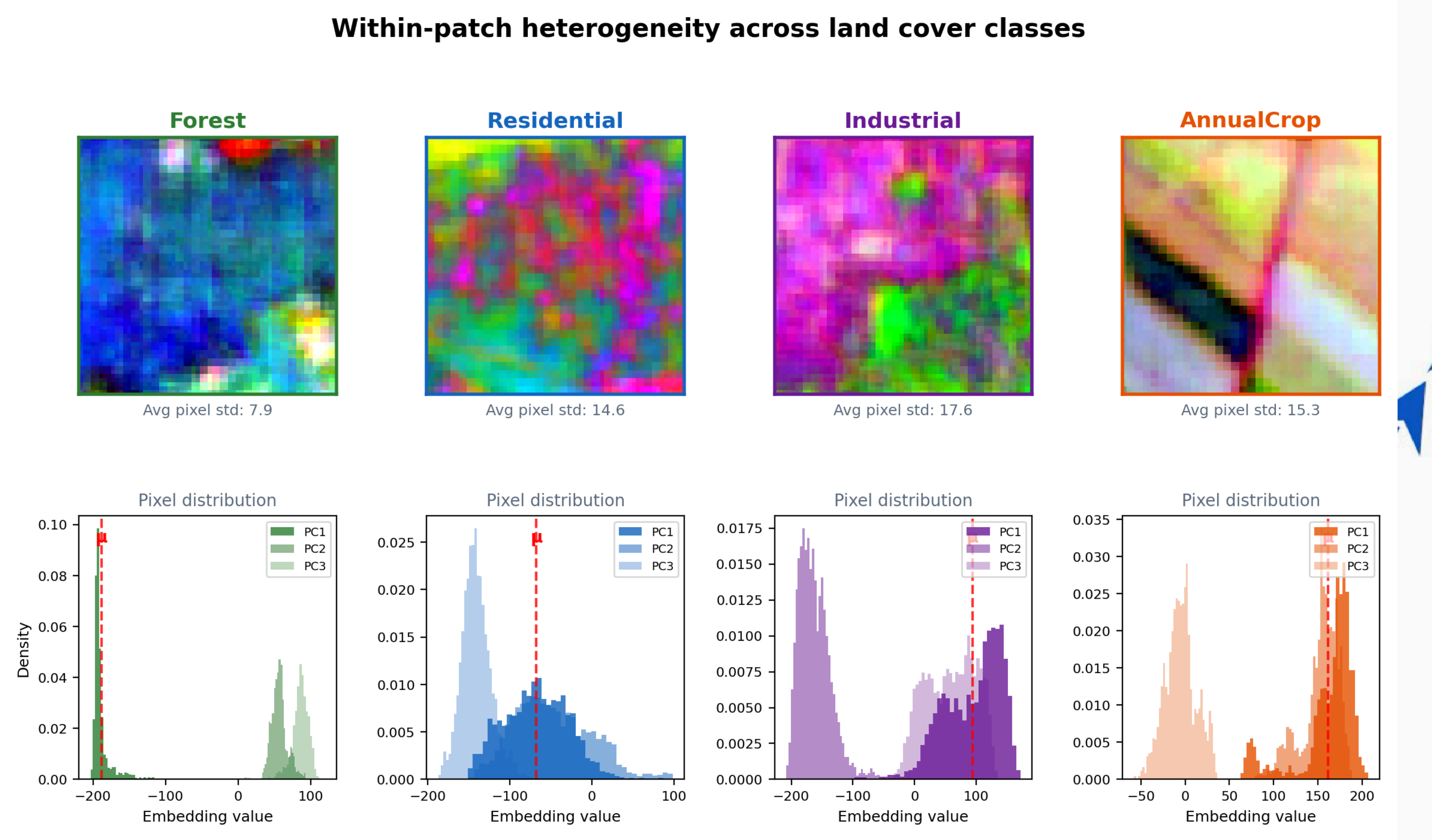
- PCA — project to lower dim
- BoVW — k-means histogram

E.g., a residential patch and an industrial area can share a mean but differ wildly in variance.

Within-patch heterogeneity across classes

Homogeneous classes cluster tightly; heterogeneous ones show **broad distributions**.

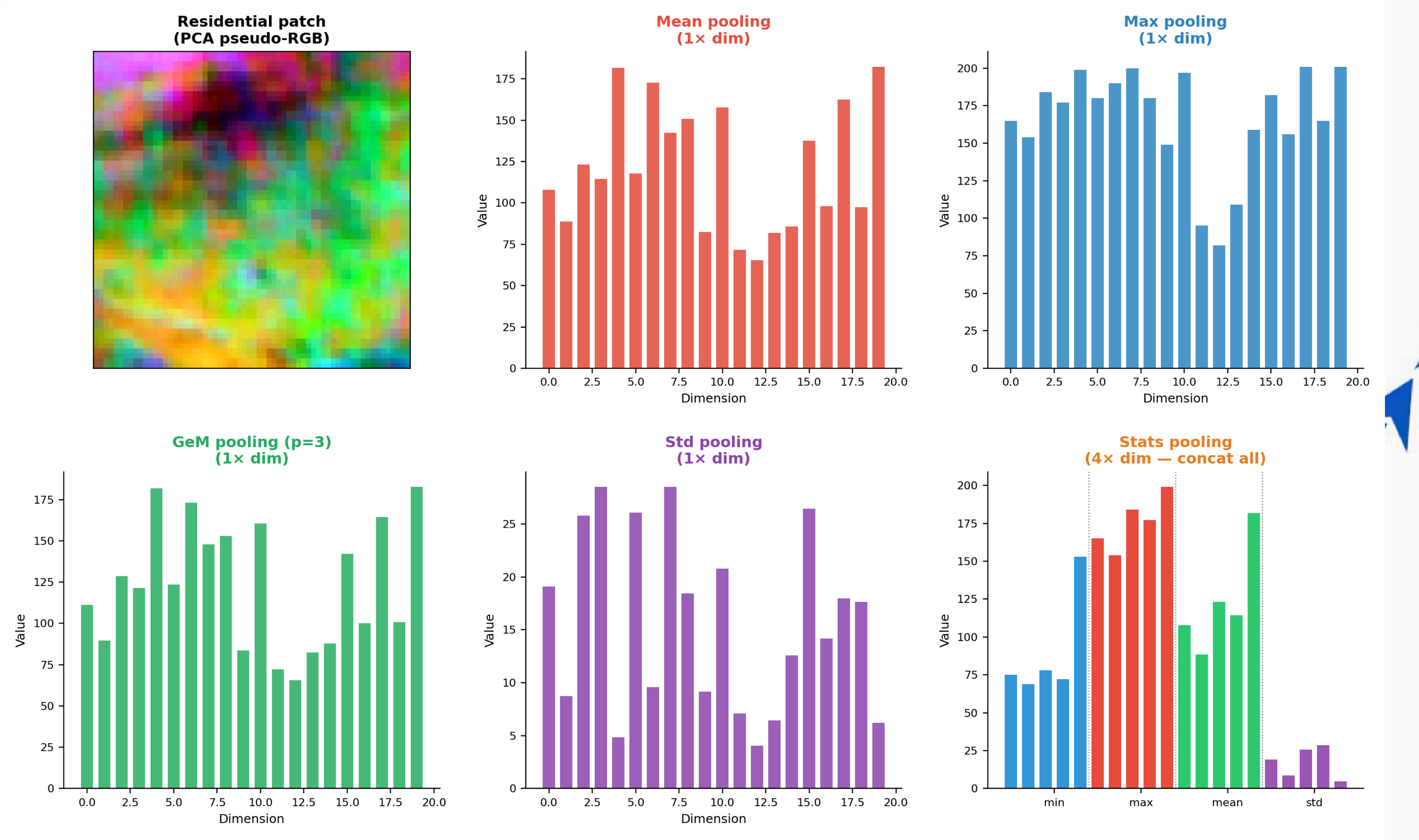
Mean pooling (μ, red) misses this signal.



What different pooling methods capture

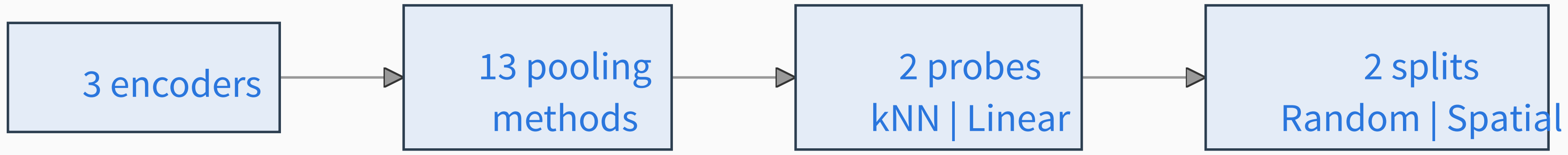
Same patch, different pooling outputs.

Stats (4× dim) captures the full distribution; mean (1×) collapses it.



Setup

Evaluation protocol



= 156 experiments

kNN (k=5, cosine distance)
Probes geometric / similarity structure

Linear (logistic regression, L-BFGS)
Probes linear separability

Results

Distributional statistics improve generalization

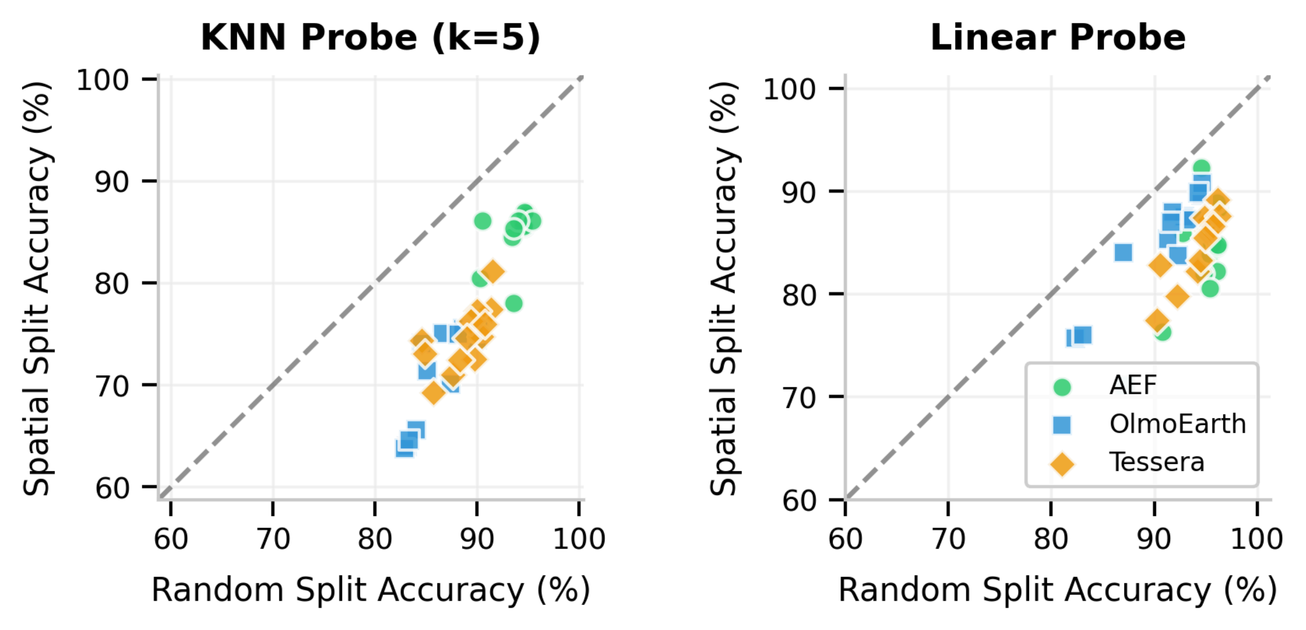
Linear probe on **spatial** splits:

Method	Accuracy
Mean (baseline)	76–84%
Mean+Max	87–89%
Stats	**88–91%**

+7–10% gain over mean pooling.

On random splits, differences shrink to 3–5% (spatial leakage flatters everyone).

The spatial split is the honest evaluation.

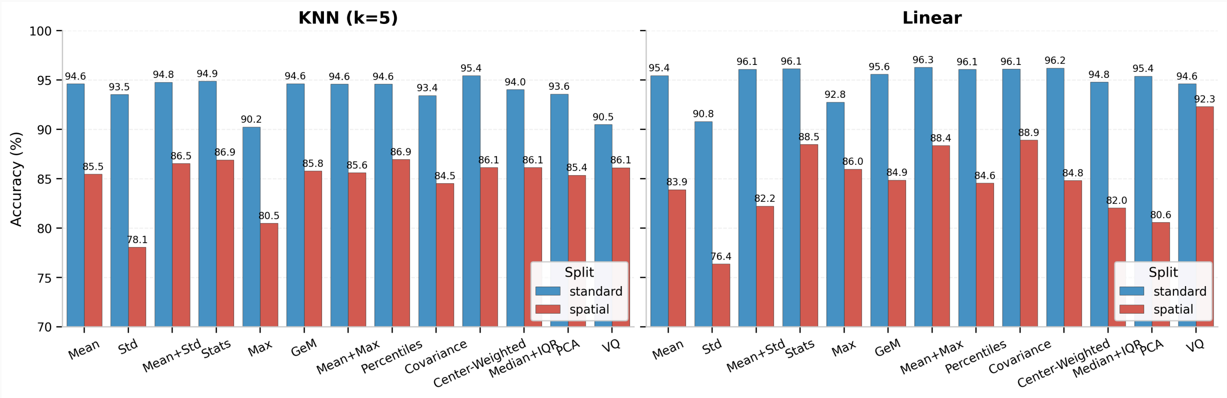


Closer to diagonal = smaller gap

GeM: zero-cost upgrade

Generalized Mean (p=3):

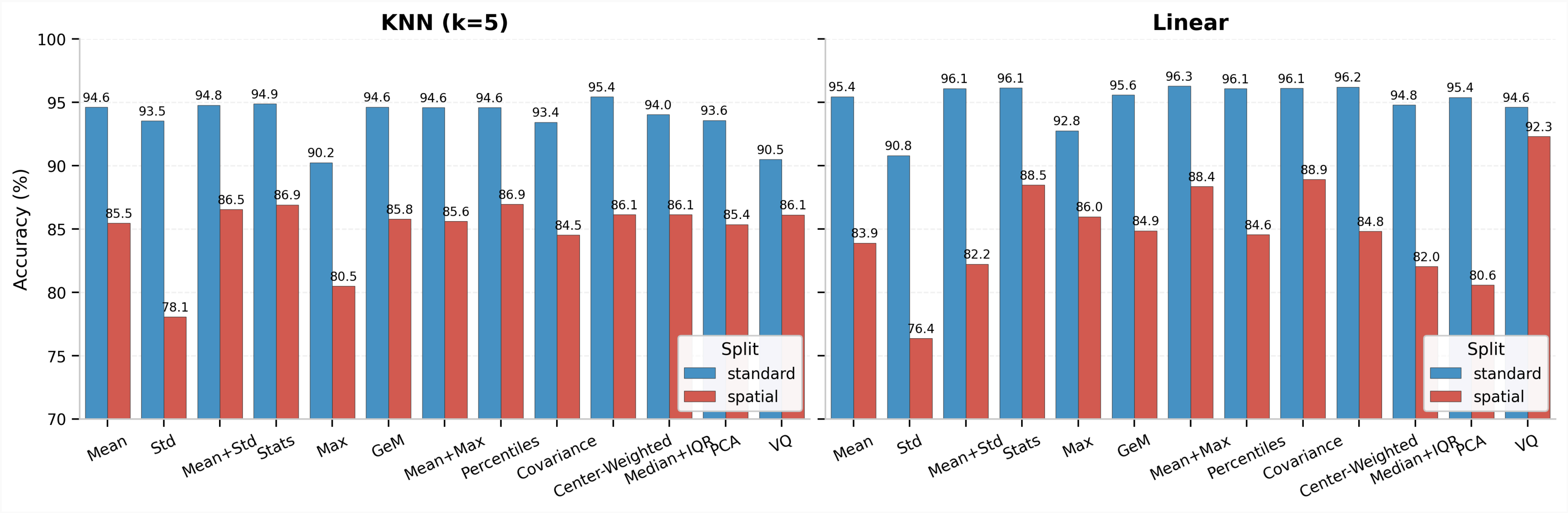
$$\text{GeM} = \left(\frac{1}{N} \sum_i x_i^{\,3} \right)^{1/3}$$



- **+5% spatial accuracy** over mean
- Same dimensionality (1×)
- No training, no storage overhead
- Consistent across all three encoders

Softly upweights high-activation pixels without the harsh cutoff of max pooling.

Accuracy–dimensionality tradeoff



No overhead (1×)
GeM (+5%)

Best tradeoff (2×)
Mean+Max (+6–8%)

Peak accuracy (4×)
Stats (+7–10%)

Pooling interacts with the encoder

AlphaEarth (64-d)

BoVW hits 92.3% (linear) but fails on other encoders.

Compact embeddings are naturally more robust to shift.

OlmoEarth (128-d)

Std alone boosts kNN by +11% over mean.

Second-order stats help similarity search more than linear probes.

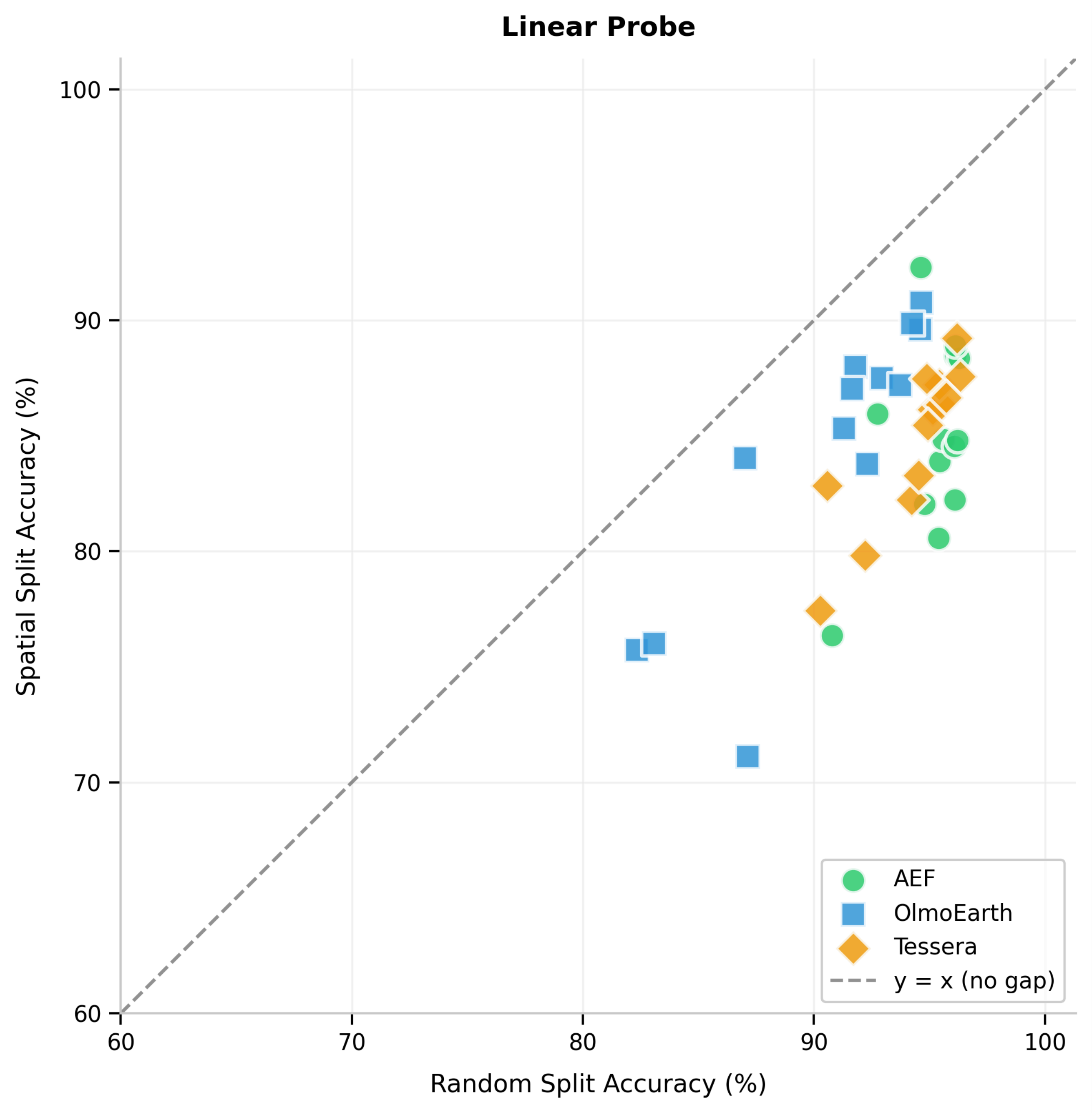
Tessera (512-d)

Covariance pooling excels at kNN.

Largest mean-pooling gap (12%). High-dim benefits most from stats.

Higher embedding dimension $\Longrightarrow$ more benefit from distributional statistics.

Richer statistics close the generalization gap



Accuracy drop (random → spatial):

Method	Drop
Mean	**10%**
Mean+Max	6.4%
Stats	6.2%
Max	5.8%

Mean pooling discards intra-patch heterogeneity that matters most under geographic shift.

Takeaways

Recommendations

Scenario	Method	Gain	Dim
Drop-in default	GeM (p=3)	+5%	1×
Best accuracy	Stats [min/max/μ/σ]	+7–10%	4×
Balanced	Mean+Max	+6–8%	2×
kNN retrieval	Covariance / Percentiles	+3–7%	varies

All recommended methods are **training-free** and consistent across encoders.

Start with GeM. Upgrade to Stats if you can afford 4× the storage.

Why does this work?

The intuition

Mean pooling collapses a patch into a single “average pixel.”

A residential neighborhood and an industrial park can share a similar mean but differ in variance — mixed rooftops, roads, vegetation vs. uniform warehouses.

Higher-order statistics preserve that heterogeneity.

Broader implications

Extends GeM results from image retrieval to the geospatial domain.

Foundation models encode fine-grained semantic structure worth preserving.

Aggregation should be a design decision, not an afterthought.

Limitations & future directions

Limitations

- Single benchmark (EuroSAT)
- Three embedding sources
- Fixed hyperparameters

Next steps

- BigEarthNet, fMoW
- Learned / attention pooling
- Temporal aggregation

Open resources

- Dataset: hf.co/datasets/isaaccorley/eurosat-embed
- Code: github.com/isaaccorley/geopool

Questions?

Code

Dataset